MVC Framework Introduction

The Model–View–Controller (MVC) framework is a design pattern that separates an application into three parts—Model, View, and Controller—making web applications easier to build, test, and maintain. ASP.NET MVC is Microsoft's implementation of this pattern, widely used for creating dynamic, scalable, and structured web apps.

What is MVC?

Model: Handles the data and business logic (e.g., database, classes, validation).
View: Responsible for the user interface (HTML, CSS, Razor templates).
Controller: Manages user requests, processes input, and decides which view to show.
This separation ensures clean code, reusability, and easier debugging.

How MVC Works

- User Request: A user enters a URL or clicks a link.
- Controller: The request goes to a controller action method.
- Model: The controller may fetch or update data using the model.
- View: The controller sends data to a view, which renders HTML.
- Response: The view is returned to the browser.

Features of ASP.NET MVC

- Separation of Concerns: Each part (Model, View, Controller) has a distinct role.
- Testability: Easy to write unit tests for controllers and models.
- Extensibility: Supports filters, routing, dependency injection.
- Cross-platform: ASP.NET Core MVC runs on Windows, Linux, and macOS.

Advantages of MVC

- Organized Code: Easier to maintain large projects.
- Reusability: Views and models can be reused across controllers.
- Scalability: Suitable for enterprise-level applications.
- Flexibility: Supports RESTful APIs and modern web standards.

Disadvantages of MVC

- Learning Curve: Beginners may find it complex at first.
- More Files: Requires multiple files (controller, model, view) for even simple tasks.
- Setup Overhead: Compared to Web Forms, initial setup is more involved.

Example in ASP.NET MVC

Controller:

```
public class HomeController : Controller
{
   public ActionResult Index()
   {
      return View();
   }
}
```

View (Index.cshtml):

```
@{
   ViewBag.Title = "Home Page";
}
<h2>Welcome to MVC!</h2>
```

When you run /Home/Index, the controller executes and returns the view.

Popular MVC Frameworks

- ASP.NET MVC / ASP.NET Core MVC (Microsoft)
- Spring MVC (Java)
- Ruby on Rails (Ruby)
- Django (Python)
- Laravel (PHP)